where it is understood that if $f(x)$ is not continuous at x the left side must be replaced by $\dfrac{f(x+0)+f(x-0)}{2}$.

These results can be simplified somewhat if $f(x)$ is either an odd or an even function, and we have

$$f(x) = \frac{2}{\pi}\int_0^\infty \sin \alpha x \, d\alpha \int_0^\infty f(u) \sin \alpha u \, du \qquad \text{if } f(x) \text{ is odd} \tag{5}$$

$$f(x) = \frac{2}{\pi}\int_0^\infty \cos \alpha x \, d\alpha \int_0^\infty f(u) \cos \alpha u \, du \qquad \text{if } f(x) \text{ is even} \tag{6}$$

FOURIER TRANSFORMS

From (4) it follows that if

$$F(\alpha) = \int_{-\infty}^\infty f(u)\, e^{-i\alpha u} \, du \tag{7}$$

then

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^\infty F(\alpha)\, e^{i\alpha x} \, d\alpha \tag{8}$$

The function $F(\alpha)$ is called the *Fourier transform* of $f(x)$ and is sometimes written $F(\alpha) = \mathcal{F}\{f(x)\}$. The function $f(x)$ is the *inverse Fourier transform* of $F(\alpha)$ and is written $f(x) = \mathcal{F}^{-1}\{F(\alpha)\}$.

Note: The constants 1 and $1/2\pi$ preceding the integral signs in (7) and (8) could be replaced by any two constants whose product is $1/2\pi$. In this book, however, we shall keep to the above choice.

FOURIER SINE AND COSINE TRANSFORMS

If $f(x)$ is an odd function, then Fourier's integral theorem reduces to (5). If we let

$$F_S(\alpha) = \int_0^\infty f(u) \sin \alpha u \, du \tag{9}$$

then it follows from (5) that

$$f(x) = \frac{2}{\pi}\int_0^\infty F_S(\alpha) \sin \alpha x \, d\alpha \tag{10}$$

We call $F_S(\alpha)$ the *Fourier sine transform* of $f(x)$, while $f(x)$ is the *inverse Fourier sine transform* of $F_S(\alpha)$.

Similarly, if $f(x)$ is an even function, Fourier's integral theorem reduces to (6). Thus if we let

$$F_C(\alpha) = \int_0^\infty f(u) \cos \alpha u \, du \tag{11}$$

then it follows from (6) that

$$f(x) = \frac{2}{\pi}\int_0^\infty F_C(\alpha) \cos \alpha x \, d\alpha \tag{12}$$

We call $F_C(\alpha)$ the *Fourier cosine transform* of $f(x)$, while $f(x)$ is the *inverse Fourier cosine transform* of $F_C(\alpha)$.

PARSEVAL'S IDENTITIES FOR FOURIER INTEGRALS

In Chapter 2, page 23, we arrived at Parseval's identity for Fourier series. An analogy exists for Fourier integrals.

If $F(\alpha)$ and $G(\alpha)$ are Fourier transforms of $f(x)$ and $g(x)$ respectively we can show that

$$\int_{-\infty}^{\infty} f(x)\,g(x)\,dx \quad = \quad \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\alpha)\,\overline{G(\alpha)}\,d\alpha \tag{13}$$

where the bar signifies the complex conjugate obtained on replacing i by $-i$. In particular, if $f(x) = g(x)$ and hence $F(\alpha) = G(\alpha)$, then we have

$$\int_{-\infty}^{\infty} |f(x)|^2\,dx \quad = \quad \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\alpha)|^2\,d\alpha \tag{14}$$

We can refer to (*14*), or to the more general (*13*), as *Parseval's identity* for Fourier integrals.

Corresponding results can be written involving sine and cosine transforms. If $F_S(\alpha)$ and $G_S(\alpha)$ are the Fourier sine transforms of $f(x)$ and $g(x)$, respectively, then

$$\int_0^{\infty} f(x)\,g(x)\,dx \quad = \quad \frac{2}{\pi}\int_0^{\infty} F_S(\alpha)\,G_S(\alpha)\,d\alpha \tag{15}$$

Similarly, if $F_C(\alpha)$ and $G_C(\alpha)$ are the Fourier cosine transforms of $f(x)$ and $g(x)$, respectively, then

$$\int_0^{\infty} f(x)\,g(x)\,dx \quad = \quad \frac{2}{\pi}\int_0^{\infty} F_C(\alpha)\,G_C(\alpha)\,d\alpha \tag{16}$$

In the special case where $f(x) = g(x)$, (*15*) and (*16*) become respectively

$$\int_0^{\infty} \{f(x)\}^2\,dx \quad = \quad \frac{2}{\pi}\int_0^{\infty} \{F_S(\alpha)\}^2\,d\alpha \tag{17}$$

$$\int_0^{\infty} \{f(x)\}^2\,dx \quad = \quad \frac{2}{\pi}\int_0^{\infty} \{F_C(\alpha)\}^2\,d\alpha \tag{18}$$

THE CONVOLUTION THEOREM FOR FOURIER TRANSFORMS

The *convolution* of the functions $f(x)$ and $g(x)$ is defined by

$$f * g \quad = \quad \int_{-\infty}^{\infty} f(u)\,g(x-u)\,du \tag{19}$$

An important theorem, often referred to as the *convolution theorem,* states that the Fourier transform of the convolution of $f(x)$ and $g(x)$ is equal to the product of the Fourier transforms of $f(x)$ and $g(x)$. In symbols,

$$\mathcal{F}\{f * g\} \quad = \quad \mathcal{F}\{f\}\,\mathcal{F}\{g\} \tag{20}$$

The convolution has other important properties. For example, we have for functions f, g, and h:

$$f * g \;=\; g * f, \quad f * (g * h) = (f * g) * h, \quad f * (g + h) \;=\; f * g + f * h \tag{21}$$

i.e., the convolution obeys the commutative, associative and distributive laws of algebra.

APPLICATIONS OF FOURIER INTEGRALS AND TRANSFORMS

Fourier integrals and transforms can be used in solving a variety of boundary value problems arising in science and engineering. See Problems 5.20–5.22.

Solved Problems

THE FOURIER INTEGRAL AND FOURIER TRANSFORMS

5.1. Show that (*1*) and (*3*), page 80, are equivalent forms of Fourier's integral theorem.

Let us start with the form

$$f(x) = \frac{1}{\pi}\int_{\alpha=0}^{\infty}\int_{u=-\infty}^{\infty} f(u)\cos\alpha(x-u)\,du\,d\alpha \tag{1}$$

which is proved later (see Problems 5.10–5.14). The result (*1*) can be written as

$$f(x) = \frac{1}{\pi}\int_{\alpha=0}^{\infty}\int_{u=-\infty}^{\infty} f(u)[\cos\alpha x\cos\alpha u + \sin\alpha x\sin\alpha u]\,du\,d\alpha$$

or

$$f(x) = \int_{\alpha=0}^{\infty}\{A(\alpha)\cos\alpha x + B(\alpha)\sin\alpha x\}\,d\alpha \tag{2}$$

where we let

$$\left.\begin{aligned} A(\alpha) &= \frac{1}{\pi}\int_{-\infty}^{\infty} f(u)\cos\alpha u\,du \\ B(\alpha) &= \frac{1}{\pi}\int_{-\infty}^{\infty} f(u)\sin\alpha u\,du \end{aligned}\right\} \tag{3}$$

Conversely, by substituting (*3*) into (*2*) we obtain (*1*). Thus the two forms are equivalent.

5.2. Show that (*3*) and (*4*), page 80, are equivalent.

We have from (*3*), page 80, and the fact that $\cos\alpha(x-u)$ is an even function of α:

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(u)\cos\alpha(x-u)\,du\,d\alpha \tag{1}$$

Then, using the fact that $\sin\alpha(x-u)$ is an odd function of α, we have

$$0 = \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(u)\sin\alpha(x-u)\,du\,d\alpha \tag{2}$$

Multiplying (*2*) by i and adding to (*1*) we then have

$$\begin{aligned} f(x) &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(u)[\cos\alpha(x-u) + i\sin\alpha(x-u)]\,du\,d\alpha \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(u)e^{i\alpha(x-u)}\,du\,d\alpha \end{aligned}$$

Similarly we can deduce that (*3*), page 80, follows from (*4*).

5.3. (*a*) Find the Fourier transform of $f(x) = \begin{cases} 1 & |x| < a \\ 0 & |x| > a \end{cases}$.

(*b*) Graph $f(x)$ and its Fourier transform for $a = 3$.

(*a*) The Fourier transform of $f(x)$ is

$$\begin{aligned} F(\alpha) &= \int_{-\infty}^{\infty} f(u)e^{-i\alpha u}\,du = \int_{-a}^{a}(1)e^{-i\alpha u}\,du = \left.\frac{e^{-i\alpha u}}{-i\alpha}\right|_{-a}^{a} \\ &= \frac{e^{i\alpha a} - e^{-i\alpha a}}{i\alpha} = 2\frac{\sin\alpha a}{\alpha}, \quad \alpha \neq 0 \end{aligned}$$

For $\alpha = 0$, we obtain $F(\alpha) = 2a$.

(b) The graphs of $f(x)$ and $F(\alpha)$ for $a = 3$ are shown in Figs. 5-1 and 5-2 respectively.

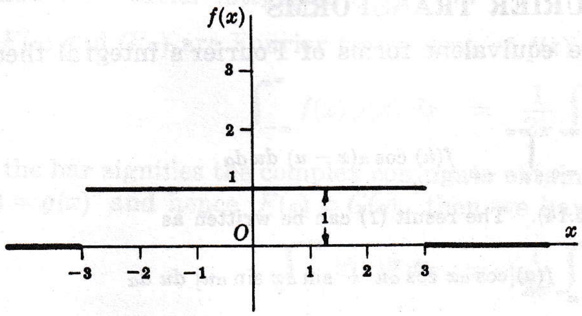

Fig. 5-1

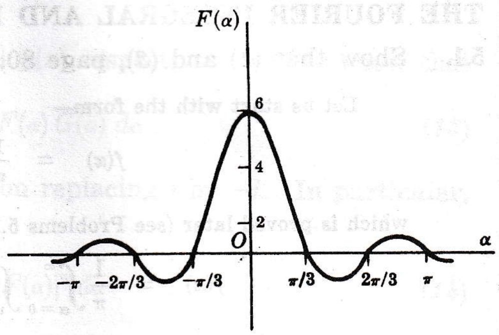

Fig. 5-2

5.4. (a) Use the result of Problem 5.3 to evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{\sin \alpha a \cos \alpha x}{\alpha}\, d\alpha$.

(b) Deduce the value of $\displaystyle\int_0^{\infty} \frac{\sin u}{u}\, du$.

(a) From Fourier's integral theorem, if

$$F(\alpha) = \int_{-\infty}^{\infty} f(u)e^{-i\alpha u}\, du \quad \text{then} \quad f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\alpha)e^{i\alpha x}\, d\alpha$$

Then from Problem 5.3,

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} 2\frac{\sin \alpha a}{\alpha} e^{i\alpha x}\, d\alpha = \begin{cases} 1 & |x| < a \\ 1/2 & |x| = a \\ 0 & |x| > a \end{cases} \tag{1}$$

The left side of (1) is equal to

$$\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\sin \alpha a \cos \alpha x}{\alpha}\, d\alpha + \frac{i}{\pi}\int_{-\infty}^{\infty} \frac{\sin \alpha a \sin \alpha x}{\alpha}\, d\alpha \tag{2}$$

The integrand in the second integral of (2) is odd and so the integral is zero. Then from (1) and (2), we have

$$\int_{-\infty}^{\infty} \frac{\sin \alpha a \cos \alpha x}{\alpha}\, d\alpha = \begin{cases} \pi & |x| < a \\ \pi/2 & |x| = a \\ 0 & |x| > a \end{cases} \tag{3}$$

(b) If $x = 0$ and $a = 1$ in the result of (a), we have

$$\int_{-\infty}^{\infty} \frac{\sin \alpha}{\alpha}\, d\alpha = \pi \quad \text{or} \quad \int_0^{\infty} \frac{\sin \alpha}{\alpha}\, d\alpha = \frac{\pi}{2}$$

since the integrand is even.

5.5. (a) Find the Fourier cosine transform of $f(x) = e^{-mx}$, $m > 0$.

(b) Use the result in (a) to show that

$$\int_0^{\infty} \frac{\cos pv}{v^2 + \beta^2}\, dv = \frac{\pi}{2\beta} e^{-p\beta} \qquad (p > 0, \beta > 0)$$

(a) The Fourier cosine transform of $f(x) = e^{-mx}$ is by definition

$$F_C(\alpha) = \int_0^\infty e^{-mu} \cos \alpha u \, du$$

$$= \left. \frac{e^{-mu}(-m \cos \alpha u + \alpha \sin \alpha u)}{m^2 + \alpha^2} \right|_0^\infty$$

$$= \frac{m}{m^2 + \alpha^2}$$

(b) From (12), page 81, we have

$$f(x) = \frac{2}{\pi} \int_0^\infty F_C(\alpha) \cos \alpha x \, d\alpha$$

or

$$e^{-mx} = \frac{2}{\pi} \int_0^\infty \frac{m \cos \alpha x}{m^2 + \alpha^2} \, d\alpha$$

i.e.

$$\int_0^\infty \frac{\cos \alpha x}{m^2 + \alpha^2} \, d\alpha = \frac{\pi}{2m} e^{-mx}$$

Replacing α by v, x by p, and m by β, we have

$$\int_0^\infty \frac{\cos pv}{v^2 + \beta^2} \, dv = \frac{\pi}{2\beta} e^{-p\beta}, \qquad p > 0,\ \beta > 0$$

5.6. Solve the integral equation

$$\int_0^\infty f(x) \sin \alpha x \, dx = \begin{cases} 1 - \alpha & 0 \leqq \alpha \leqq 1 \\ 0 & \alpha > 1 \end{cases}$$

If we write

$$F_S(\alpha) = \int_0^\infty f(x) \sin \alpha x \, dx = \begin{cases} 1 - \alpha & 0 \leqq \alpha \leqq 1 \\ 0 & \alpha > 1 \end{cases}$$

then, by (10), page 81,

$$f(x) = \frac{2}{\pi} \int_0^\infty F_S(\alpha) \sin \alpha x \, d\alpha$$

$$= \frac{2}{\pi} \int_0^1 (1 - \alpha) \sin \alpha x \, d\alpha$$

$$= \frac{2(x - \sin x)}{\pi x^2}$$

THE CONVOLUTION THEOREM

5.7. Prove the convolution theorem on page 82.

We have by definition of the Fourier transform

$$F(\alpha) = \int_{-\infty}^\infty f(u) e^{-i\alpha u} \, du, \qquad G(\alpha) = \int_{-\infty}^\infty g(v) e^{-i\alpha v} \, dv \tag{1}$$

Then

$$F(\alpha)\, G(\alpha) = \int_{-\infty}^\infty \int_{-\infty}^\infty f(u)\, g(v) e^{-i\alpha(u+v)} \, du \, dv \tag{2}$$

Let $u + v = x$ in the double integral (2) which we wish to transform from the variables (u, v) to the variables (u, x). From advanced calculus we know that

$$du \, dv = \frac{\partial(u, v)}{\partial(u, x)} \, du \, dx \tag{3}$$